Shoug Alomran

Network Topologies

Feature	Mesh	Star	Bus	Ring	Hybrid
Basic shape	Every device connects to **every other device**	Every device connects to **one central controller**	All devices connect to **one main cable**	Each device connects to the **two devices beside it**	Combination of multiple topologies
Connection type	Dedicated point-to-point	Dedicated point-to-point to central device	Shared main link	Point-to-point between neighbors	Depends on combination
Number of links	**N(N−1)/2**	**N**	**1 main + N drop lines**	**N**	Depends on design
Cabling	🔴 Most cabling	🟡 Less than mesh	🟢 Less cabling	🟢 Less cabling	Depends
Installation	🔴 Difficult	🟢 Easy	🟢 Easy	🟢 Easy	Depends
Add/remove devices	Difficult due to many connections	🟢 Easy	🔴 Difficult	🟢 Easy	Depends
Fault isolation	🟢 Easy	🟢 Easy	🔴 Difficult	🟢 Easy	Depends
If one ordinary link fails	Network can still work	Network can still work	A break in the bus can stop transmission	Ring is **not robust**	Depends
Major weakness	Expensive + lots of wiring	**Central device failure stops network**	**Main bus cable failure stops transmission**	**Not robust**	Depends on component topologies
Security	🟢 Secure	🟡 Secure, but less than mesh	Not specified in slides	Not specified in slides	Not specified in slides
Main advantage	Very robust, secure, no traffic problems	Easy, cheaper, robust	Simple and uses less cabling	Simple, less cable, easy fault isolation	Can combine different topology structures

The easiest way to distinguish them

Mesh = EVERYONE connects to EVERYONE

```
A— B
|\   /|
| \ / |
| / \ |
|/   \|
C—D
```

Star = EVERYONE connects to the CENTER

```
   A
   |
B — HUB — C
   |
   D
```

Bus = EVERYONE connects to ONE LINE

```
____________________
 |    |    |
 A    B    C
```

Ring = EVERYONE forms a CIRCLE

```
A ——— B
|         |
D ——— C
```

Hybrid = MIX different topologies

For example, the slide shows a **star backbone connected to bus networks**.

The formulas you should know

For **N devices**:

Mesh

Links= $\frac{N(N-1)}{2}$

and each device needs: $N - 1\, ports$

Star

Links= N

Ring

Links= N

Bus

$1 Main\ Link\ +\ N\ Drop\ Lines$